



INTRODUCTION TO DIGITAL SIGNAL PROCESSING (1)

Notes By

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WHAT IS DIGITAL SIGNAL PROCESSING (DSP)?

Digital Signal Processing is the process of representing signals in a discrete mathematical sequence of numbers and analyzing, modifying, and extracting the information contained in the signal by carrying out algorithmic operations and processing on the signal.

DSP BREAKDOWN

DIGITAL - means discrete in nature – i.e. the signal levels are chosen from a finite set of levels, as opposed to continuous or analog signals, which can have an infinite number of levels. In practice, digital nearly always means binary, that is, two-level signals – our standard TTL or CMOS levels as used in computers and other digital systems. Note that signals used in DSP systems may be developed from analog signals by sampling and analog-to-digital conversion or may be available as digital signals initially, as from another digital system.

DSP BREAKDOWN

DSP signals are also discrete in time, i.e. they represent samples taken at specific instants in time. Thus, we use notation like $x[n]$ or $y[n]$ to represent these signals, where n is an integer that represents, effectively, the sample number.

DSP BREAKDOWN

PROCESSING – refers to the applications we want to implement or operations we want to perform on the digital signal. The two major, end-result applications for digital signal processing are digital filters and the fast Fourier transform (FFT). However, there are innumerable other applications or types of processing, carried out because they are important in themselves or because they are steps in implementing filters or FFTs.

EXERCISE

What is FFT?

Why is it important in DSP?

Give areas of application of FFT.

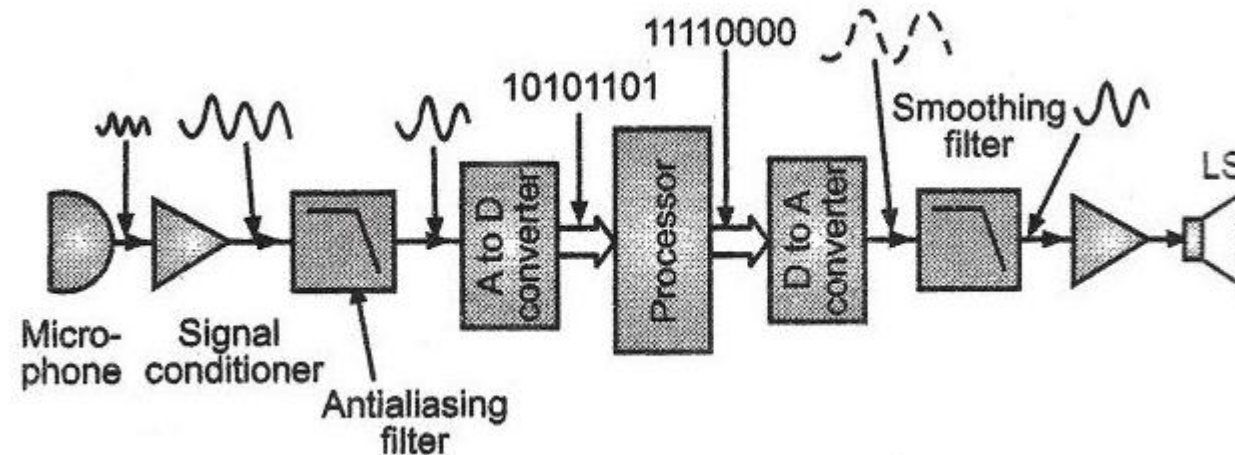
WHAT IS A SIGNAL?

SIGNAL – means some physical quantity whose variations convey information. A signal can be mechanical, hydraulic, pneumatic, optical (visible, UV or infra-red light), temperature, etc. However, we generally deal with electrical signals, either because they were initially developed as electrical, or because they have been converted to electrical.

BASIC ELEMENT OF A DSP SYSTEM



General Block Diagram of a DSP System



DSP STEP BY STEP

- ✓ The first step is to get an electrical signal. The transducer (in the previous diagram, a microphone) converts sound into an electrical signal. Any transducer can be used
- ✓ Once we have an analog electrical signal, we pass it through an operational amplifier (Op-Amp) to condition the analog signal. Basically, we amplify the signal. Or limit it to protect the next stages. Signal conditioning is the manipulation of a signal in a way that prepares it for the next stage of processing.

DSP STEP BY STEP

- ✓ The anti-aliasing filter is an essential step in the conversion of analog to a digital signal. It is a low-pass filter. Meaning, it allows frequencies up to a certain threshold to pass. It attenuates all frequencies above this threshold. These unwanted frequencies make it difficult to sample an analog signal.
- ✓ The next stage is a simple analog-to-digital converter (ADC). This unit takes in analog signals and outputs a stream of binary digits.

DSP STEP BY STEP

- ✓ The heart of the system is the digital signal processor. These days we use CMOS chips (even ULSI) to make digital signal processors. In fact, modern processors, like the Cortex M4 have DSP units built inside the SoC (System on Chip). These processor units have high-speed, high data throughputs, and dedicated instruction sets.

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DSP OUTPUT

- ✓ The digital-to-analog converter does what its name implies. It's necessary for the slew rate of the DAC to match the acquisition rate of the ADC.
- ✓ The smoothing filter is another low-pass filter that smoothens the output by removing unwanted high-frequency components.
- ✓ The last op-amp is just an amplifier.
- ✓ The output transducer is a speaker in this example

EXERCISE

What is Slew Rate?

Why does the slew rate of the DAC have to match the acquisition rate of the ADC?

APPLICATIONS OF A DIGITAL SIGNAL PROCESSING SYSTEM

1. Telecommunication

For echo cancellation.

Equalization – Think about tuning your radio for bass and treble).

Filtering – Removing unwanted signals using specially designed filters like the Infinite Impulse Response Filter (IIR).

Multiplexing and repeating signals.

APPLICATIONS OF A DIGITAL SIGNAL PROCESSING SYSTEM

2. Instrumentation and Control

In designing Phase Locked Logic (PLL).

Noise reduction circuits.

Compression of signals.

Function generators.

3. Digital Image Processing

Compression of an image.

APPLICATIONS OF A DIGITAL SIGNAL PROCESSING SYSTEM

4. Enhancement, reconstruction, and restoration of an image.

Analysis or face detection (like Snapchat).

5. Speech Processing

Digital audio synthesis.

Speech recognition and analysis.

APPLICATIONS OF A DIGITAL SIGNAL PROCESSING SYSTEM

6. Medicine

X-rays, ECGs, EEGs.

7. Signal filtering

Noise removal and shaping of signal spectrums.

APPLICATIONS OF A DIGITAL SIGNAL PROCESSING SYSTEM

8. Military

Sonar and navigation.

Analysis after tracking in radars.

9. Consumer electronics

Music players

Professional music turntables (like the ones DJs use).

ADVANTAGES OF DIGITAL SIGNAL PROCESSING (DSP) SYSTEM

The main advantage of Digital Signal Processing is its simplicity compared to analog Signal Processing. Digital Signal Processing can perform complicated signal analysis with relative ease without the less complicated circuits.

Also, processing can be modified easily with simple changes in software, connoting flexibility.

In DSP we have access to many error detection and correction features: for example parity generation and correction.

ADVANTAGES OF DIGITAL SIGNAL PROCESSING (DSP) SYSTEM

Data storage is also much easier.

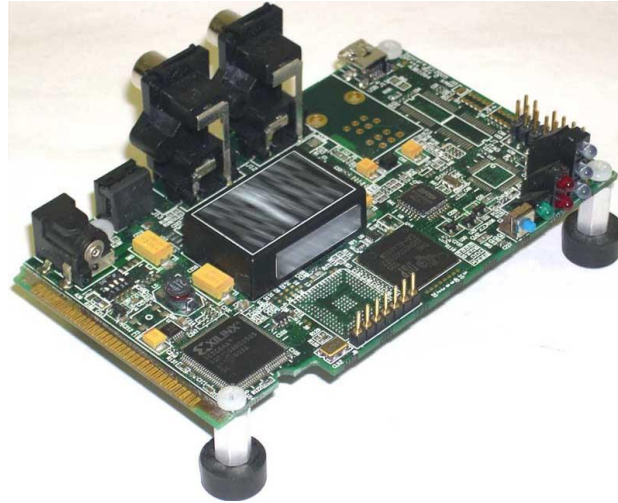
Since digital storage devices are becoming cheaper by the day, the storage of data is much easier digital forms.

It is also easier to transport and recreate digital data with 100% fidelity.

DISADVANTAGES OF DIGITAL SIGNAL PROCESSING (DSP) SYSTEM

- DSPs are made out of a large number of transistors and these transistors together consume a lot more power than analog signal processors.
- DSP has a higher learning curve required for the operation of DSP Systems.

DISADVANTAGES OF DIGITAL SIGNAL PROCESSING (DSP) SYSTEM



- Each DSP possesses different hardware architecture and software instructions, due to these issues highly skilled engineers are needed to program devices and proper training on DSP is required to program various applications.

EXERCISE

- What are filters in digital signal processing?
- What are the types of filters in digital signal processing?
- What are the disadvantages of digital signal processing?

CHARACTERISTICS OF DSP SYSTEMS

Linearity - Linearity dictates that for a single input, the output is proportional to the input, and for two or more inputs, the output must be the sum of the individual responses of the two inputs. Mathematically, this is expressed as:

if $x_1[n]$ produces $y_1[n]$, and $x_2[n]$ produces $y_2[n]$,

then $ax_1[n] + bx_2[n]$ produces $ay_1[n] + by_2[n]$

Linearity is the basis for the concept of superposition and for convolution which we will see shortly.

Time-invariant - We assume the system properties do not vary with time (or at least over the time period we are concerned about.)

CHARACTERISTICS OF DSP SYSTEMS

Causality - Causality implies that output changes do not occur before input changes. Although this cannot happen in a real situation, non-causal considerations sometimes arise in theoretical derivations. If all the samples for a certain signal have been collected and stored, causality is not an issue, since all samples both before and after a selected value of n are available. In addition, any non-causal signal can be made causal just by delaying all the samples by an appropriate amount.

CHARACTERISTICS OF DSP SYSTEMS

Most real signals and systems we deal with are causal, but just as an example, we will look at one non-causal system. To that end, consider the moving-average system, frequently used to smooth a set of data points.

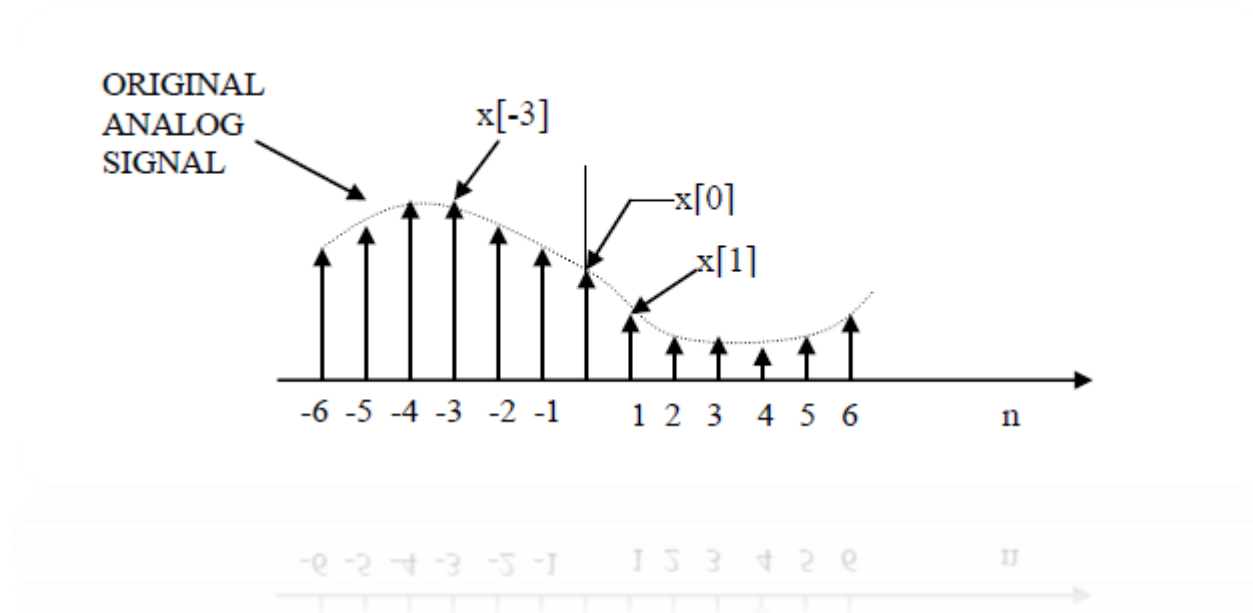
The moving average system computes the average of a certain number of points around a specific point, then replaces the value at that point with the average. Thus, a typical moving average computation might be to take the average of the two points prior to a certain point, plus the point itself, and the two points past the point in question.

SAMPLING OF ANALOGUE SIGNAL

Let us now consider the nature of the signals used in digital signal processing. These signals are frequently, but not always, developed by sampling a continuous-time signal. Sometimes they are called discrete-time signals, to reflect the fact that they have meaning only at discrete points in time. Sampled signals are also discrete in amplitude as well as time, since the normal analog to digital conversion process is finite in its resolution, and thus forces the amplitude to be chosen from a finite set of values.

SAMPLING OF ANALOGUE SIGNAL

If we call the sampled signal $x[n]$, to indicate that the values of the signal are a function of the sample number or index “ n ”, then the individual values of the signal are represented by $x[0]$, $x[1]$, $x[2]$, etc., as shown by the following:



SAMPLING OF ANALOGUE SIGNAL

Note that the index n can be negative as well as positive, to indicate that the signal sample was taken before the point designated $n=0$. This is not really a complication, since the definition of $n=0$ is frequently arbitrary.

Since the digital signals represent samples of continuous-time signals, taken at discrete points in time, they are actually a set of numbers representing the values of the continuous-time signal at the instants in time at which it was sampled. For example, typical sequences might be:

1,2,3,4,5 or 2, -4, -6, 8 or 0.330, -1.25, 5.69, 3.45, -5.77 etc.

SAMPLING OF ANALOGUE SIGNAL

Thus we have to get accustomed to the idea that the digital signals we deal with are just a string or sequence of numbers. In fact, they are usually referred to as sequences instead of signals. This string or sequence of numbers is handled just like any other set of numeric data: it can be treated as a vector, stored in an array format, manipulated by a spreadsheet, etc. A sequence is frequently called a vector, although it has no physical meaning like a force vector or electromagnetic field vector does.

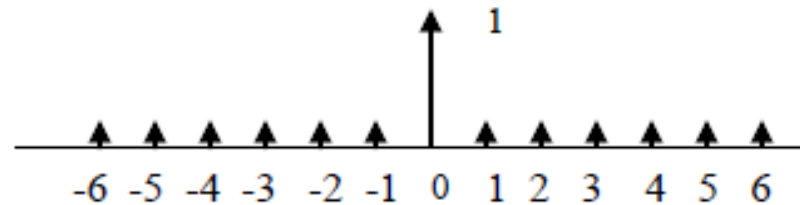
SAMPLING OF ANALOGUE SIGNAL

It is also quite possible that the sequence of numbers was not derived by sampling a continuous-time signal. They may represent data collected from measurements made in some physical or natural system, such as traffic flow, the growth rate of trees, ocean water temperature, etc., which actually are sampled quantities, but may be the result of intermediate calculations.

SAMPLING OF ANALOGUE SIGNAL

There are several special sequences used in DSP that are not derived from the sampling of a real-world signal. Rather, they are defined instead of sampled and thus have somewhat different notations.

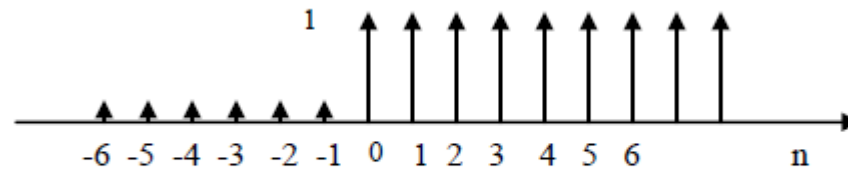
The first is the unit impulse. It is designated as $\delta[n]$, and defined as 1 for $n=0$, and 0 elsewhere. Graphically, it is $\Rightarrow$



SAMPLING OF ANALOGUE SIGNAL

The unit impulse is particularly important in DSP because DSP signals, being just sequences of numbers, can be represented by a series of impulses, each weighted to represent the actual value of the sequence for each value of n . Thus, DSP system analysis frequently amounts to analysis of the system response to a sequence of impulses individually, and subsequent superposition or addition of the individual responses to find the complete response to the input sequence.

SAMPLING OF ANALOGUE SIGNAL



The unit step $u[n]$

Another useful defined sequence is the unit step. It is denoted by $u[n]$, and consists of a sequence of impulses with value of one for $n \geq 0$, and 0 otherwise

SAMPLING OF ANALOGUE SIGNAL

There are several other functions that can be defined, including the exponential sequence and the sinusoidal sequence. The names are more or less self-explanatory.

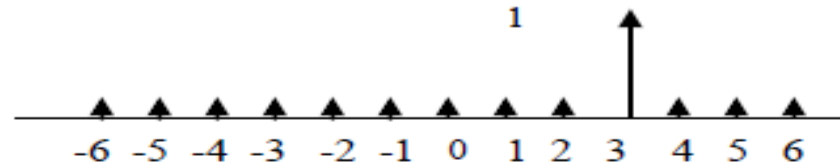
Note that there is one major difference in the notation for signals, such as $x[n]$ or $y[n]$, and defined functions such as the unit impulse $\delta[n]$ or unit step $u[n]$. For a signal such as $x[n]$, there is, in general, a numerical value associated with every value of n . Thus, $x[0]$ has a distinct value, as does $x[5]$, $x[10]$ and $x[-4]$, etc.

SAMPLING OF ANALOGUE SIGNAL

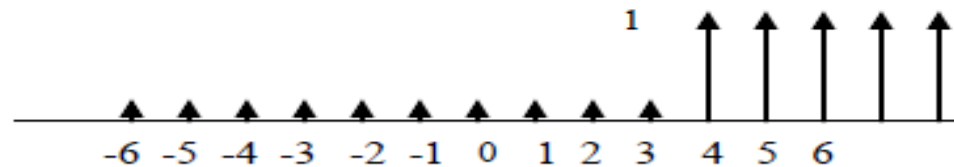
However, for a defined function such as $u[n]$, its values are determined by its definition, i.e. $d[n] = 1$ for $n=0$, and 0 elsewhere; $u[n] = 1$ for $n \geq 0$, and 0 elsewhere. It would not be correct to speak of $d[5]$ or $u[-4]$ or any other value of n . The values are determined by the definition, not by defining individual samples.

SAMPLING OF ANALOGUE SIGNAL

We can, however, talk about shifted functions. For example, $d[n-3] = 1$ for $n = 3$, and 0 elsewhere, and $u[n-4]$ is a unit step that starts at $n = 4$. Some illustrations are:



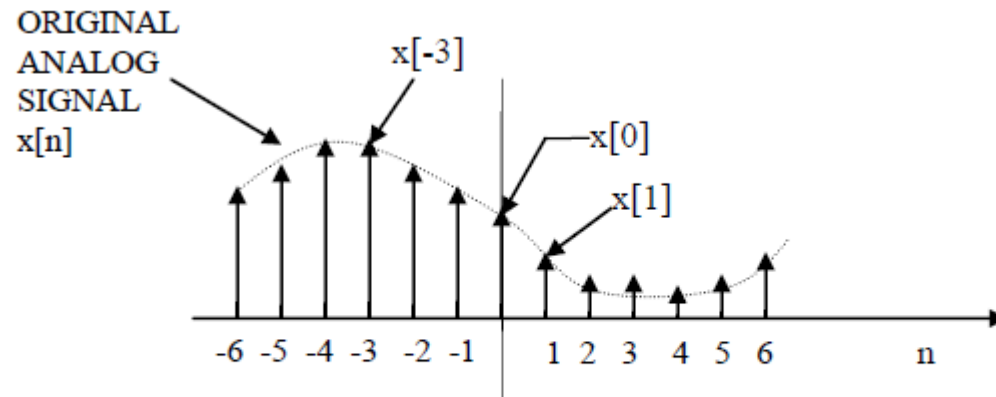
The delayed unit impulse, $\delta[n-3]$



The delayed unit step $u[n-4]$

SAMPLING OF ANALOGUE SIGNAL

A similar shifting concept applies to signals. The notation $x[n-3]$ represents signal $x[n]$ delayed by 3 units, and $y[n+4]$ represents the signal $y[n]$ advanced by 4 units. Consider, for example, the signal we used at the beginning of this section:



SAMPLING OF ANALOGUE SIGNAL

Relationships between the unit step and unit impulse. The unit step can be written in terms of the unit impulse. Since the step is really represented by an infinite sequence of impulses starting at 0, the step can be written as:

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] \text{ i.e. a sequence of impulses from 0 to infinity}$$

SAMPLING OF ANALOGUE SIGNAL

Similarly, the unit impulse can be expressed as:

$$\delta[n] = u[n] - u[n-1]$$

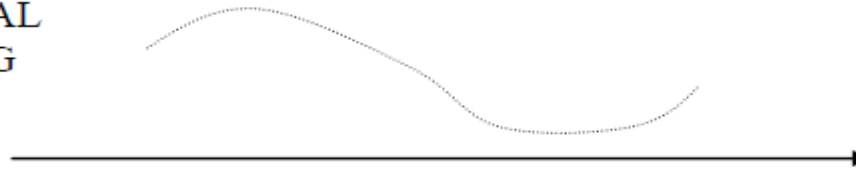
i.e. a unit step less a unit step delayed by one. This form of equation is known as a first-order difference equation. It is analogous to the differential equation used in the analysis of continuous-time systems. As we will see shortly, the difference equation is key in the modeling and analysis of DSP systems.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

First, we note that for analysis purposes it is convenient to model the conversion process as the multiplication of the analog signal by an impulse train. Since each impulse has a value of unity, the individual samples generated by this process will have the value of the analog signal at the instant that the impulse occurred. The diagram we used initially to illustrate the derivation of a digital signal from an analog signal is also useful here:

MULTIPLICATION OF AN ANALOG SIGNAL BY AN IMPULSE TRAIN

ORIGINAL
ANALOG
SIGNAL

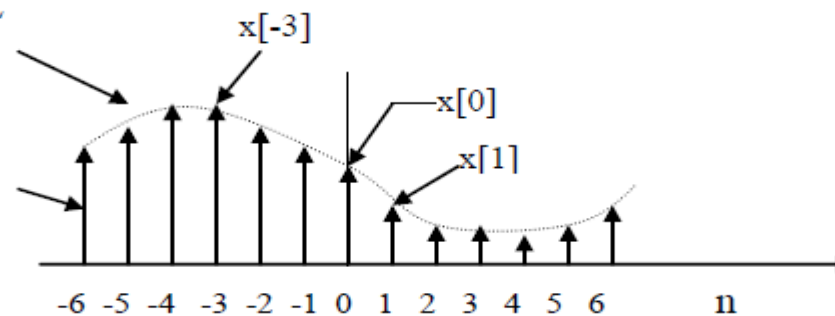


SAMPLING
SIGNAL



ORIGINAL
ANALOG
SIGNAL

SAMPLES



SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

We can see from the diagram that the multiplication of the analog signal by a train of unit-valued impulses will produce the sequence of weighted impulses shown.

The fact that the two signals are effectively multiplied allows us to use some traditional signal processing concepts to describe the results. Recall from a previous theory that multiplying two single-frequency signals together produces a result containing the sums and differences of the two original frequencies. This is confirmed by a trig identity which states that:

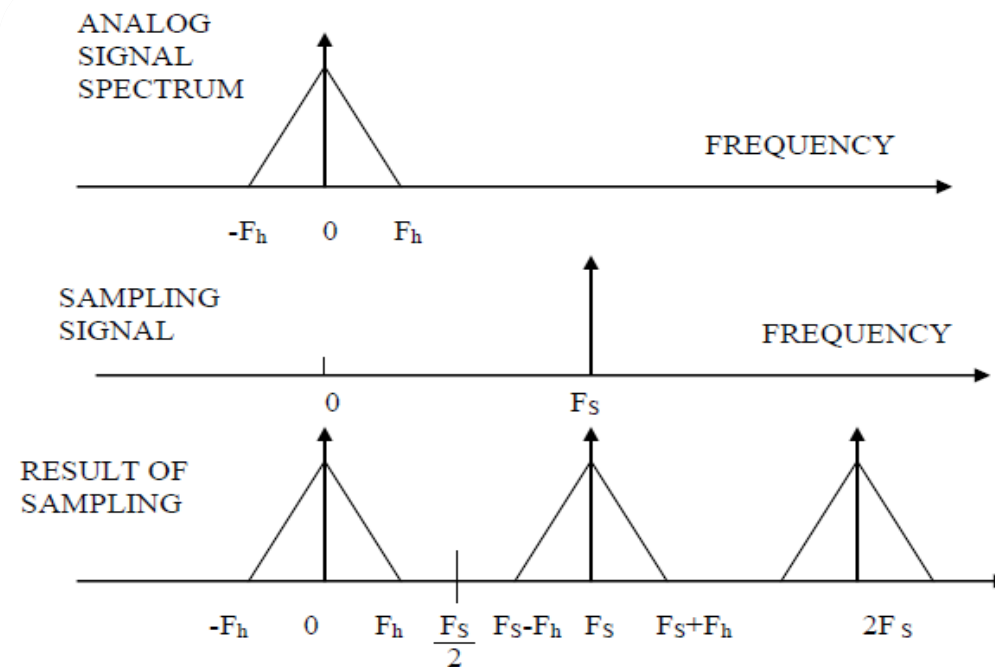
$$2\sin(a)\sin(b) = \sin(a+b) + \sin(a-b)$$

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

Since the impulse function contains its fundamental frequency, as well as all of its harmonics, the sampling operation will produce the spectrum of the analog signal centered about all the harmonics of the frequency of the impulse. The frequency of the impulse function is called the sampling frequency.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

The result of sampling an analog signal having a given spectrum, in which the highest frequency is f_h , is shown below:



SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

In effect, copies of the original spectrum appear at all multiples of the sampling frequency. With a sharp filter, it would be possible to recover all the original information just by separating out the basic spectrum from all the copies.

$$F_S - F_h > F_h \text{ so } F_S > 2F_h.$$

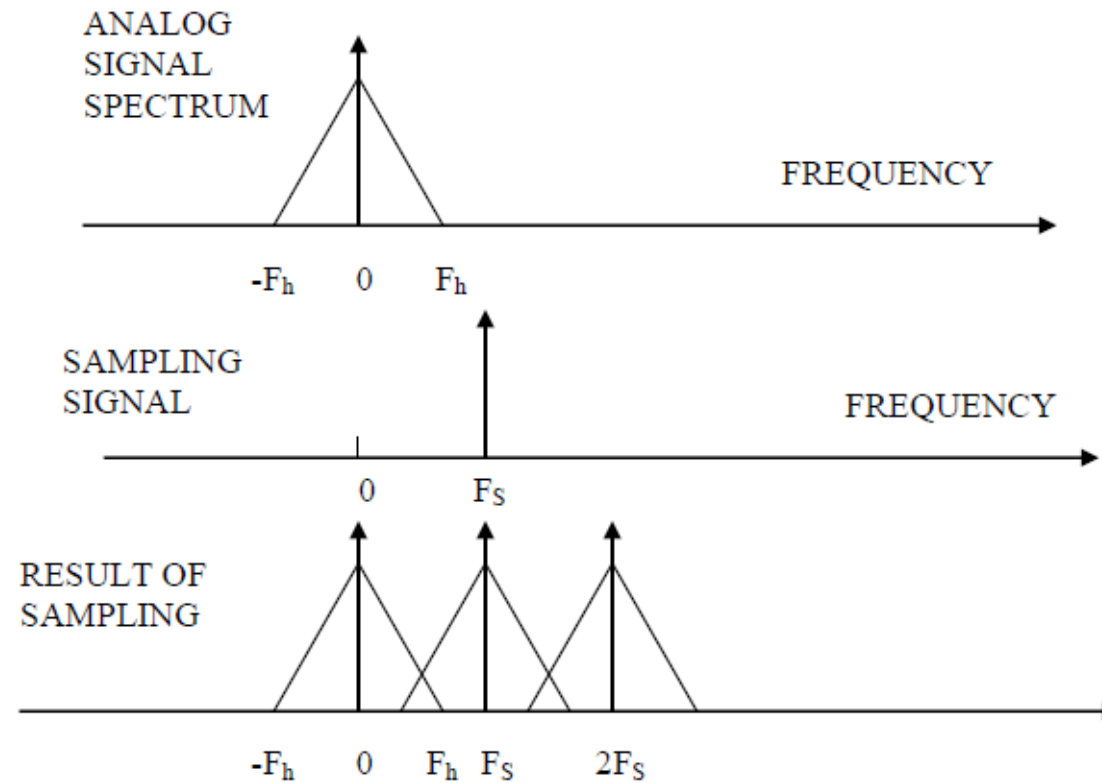
Note that in this case, the sampling frequency F_S is more than twice the highest frequency in the signal, F_h . This is obvious from the fact that

When this is the case, the original signal can be filtered out from the sampled spectrum. So, although the samples effectively contain all the copies of the original spectrum, that spectrum can be recovered without distortion.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

We use the criterion that the original spectrum can be recovered by a sharp filter as a justification for a certain sampling rate. This does not imply that we really plan to recover the original spectrum. If that's all we did, we haven't really accomplished anything. The real reason for using such a criterion is that if it is satisfied, we know that the signal is correctly represented by the sampled signal, and that it isn't distorted by the alias, which we discuss next.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION



ALIASING AND ANALOG-TO-DIGITAL CONVERSION

It is obvious from the figure that some energy from the first copy of the spectrum is injected into the original spectrum, so that even sharp filtering cannot remove that energy from the original spectrum while keeping all the information in the original signal.

This phenomenon is called aliasing, and the unwanted portion of the first copy of the spectrum that is aliased into the spectrum of the original signal is referred to as the alias signal.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

The presence of the aliasing problem means that the sampling frequency must be at least twice that of the highest frequency in the signal being sampled. This can be accomplished in two ways:

1. The original signal can be low-pass filtered to reduce the bandwidth to less than half of the sampling frequency. There are obviously limits on how far the bandwidth can be reduced and still retain the desired signal characteristics. Sometimes the filter can be a very simple RC filter, as for instance, when the signal is greatly oversampled – say at a rate 5 to 10 times F_h . In other situations, where oversampling is either impossible or not convenient, more complex filters will be required. In such cases, attention must be paid to the phase characteristic as well as the amplitude characteristics of the filter. Non-uniform phase shift, which is characteristic of many active filters, can be especially troublesome for data signals.

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

2. The sampling frequency can be increased to more than twice the highest signal frequency. There are obvious limits on this approach also, as the circuitry involved must be able to handle the sampling frequency involved.

In practice, a sampling rate of 2.5 to 3 times the highest frequency in the signal is actually used, because any filters used to recover the original signal are not ideal, and allowance must be made for that characteristic.

NYQUIST THEOREM

Nyquist Theorem The requirement that the sampling rate exceed twice F_h , the highest frequency in the signal, is known as the Nyquist Theorem or the Nyquist criterion. The frequency F_h is called the Nyquist frequency by some authors, and the frequency $2F_h$ that must be exceeded by the sampling rate is called the Nyquist rate. Others, including MATLAB, define the Nyquist Frequency as half the sampling rate.

(To be continued...)

REFERENCES

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<https://www.fmradiobroadcast.com/article/detail/Introduction-to-dsp-digital-signal-processing.html>

What is digital signal processing (DSP)? – A complete overview

Umair Hussaini | Published January 10, 2020 | Updated June 2, 2020

<https://technobyte.org/dsp-advantages-disadvantages-block-diagram-applications/>

INTRODUCTION TO DIGITAL SIGNAL PROCESSING

James Hahn, 2011, Washington State University